

1.9: Further Calculus

1. Improper Integrals

An integral is **improper** if either (a) a limit is $\pm\infty$, or (b) the integrand is unbounded at a point in the interval.

Evaluate by replacing the problem point by a parameter, then taking a limit:

$$\int_a^\infty f(x) dx = \lim_{R \rightarrow \infty} \int_a^R f(x) dx$$

$$\int_0^1 \frac{dx}{\sqrt{x}} = \lim_{\varepsilon \rightarrow 0^+} \int_\varepsilon^1 x^{-1/2} dx = \lim_{\varepsilon \rightarrow 0^+} [2\sqrt{x}]_\varepsilon^1 = 2.$$

If the limit is finite, the integral **converges**; otherwise it **diverges**.

Key p -Integrals

Integral	Converges when
$\int_1^\infty x^{-p} dx$	$p > 1$
$\int_0^1 x^{-p} dx$	$p < 1$

OCR Exam Tip: Always set up with a limit and show the work — simply writing $\int_0^\infty$ and substituting ∞ directly loses marks.

2. The Mean Value of a Function

The mean (average) value of f on $[a, b]$ is

$$\bar{f} = \frac{1}{b-a} \int_a^b f(x) dx.$$

Mean of $f(x) = \sin x$ on $[0, \pi]$:

$$\bar{f} = \frac{1}{\pi} \int_0^\pi \sin x dx = \frac{1}{\pi} [-\cos x]_0^\pi = \frac{1}{\pi} (1 + 1) = \frac{2}{\pi}.$$

3. Integration by Partial Fractions

Once a rational function is split into partial fractions (see A-level Pure), each piece integrates as:

- $\int \frac{A}{ax+b} dx = \frac{A}{a} \ln |ax+b| + C$
- $\int \frac{A}{(ax+b)^n} dx = \frac{A}{a(1-n)(ax+b)^{n-1}} + C$ (for $n \neq 1$)
- For $\frac{Ax+B}{x^2+c^2}$, split into $\frac{A}{2} \ln |x^2+c^2|$ (from d/dx of denominator) + $\frac{B}{c} \arctan(x/c)$.

4. Integration Using Trig Substitution

Integrand contains	Substitution	Derivation
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$	$dx = a \cos \theta d\theta$
$\frac{1}{a^2 + x^2}$	$x = a \tan \theta$	$dx = a \sec^2 \theta d\theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta$ or $x = a \cosh u$	Hyperbolic often easier

Standard Results (Formula Booklet)

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \arcsin\left(\frac{x}{a}\right) + C$$

$$\int \frac{dx}{a^2 + x^2} = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C$$

$$\int \frac{dx}{\sqrt{x^2 + a^2}} = \operatorname{arsinh}\left(\frac{x}{a}\right) + C$$

$$\int \frac{dx}{\sqrt{x^2 - a^2}} = \operatorname{arcosh}\left(\frac{x}{a}\right) + C \quad (\text{for } x \geq a)$$

5. Reduction Formulae

A **reduction formula** expresses an integral I_n (depending on integer n) in terms of I_{n-1} or earlier values, allowing successive calculation.

Usually derived by **integration by parts**.

Derive a reduction formula for $I_n = \int_0^{\pi/2} \sin^n x \, dx$.

Let $u = \sin^{n-1} x$, $dv = \sin x \, dx$, so $du = (n-1) \sin^{n-2} x \cos x \, dx$, $v = -\cos x$.

$$I_n = [-\sin^{n-1} x \cos x]_0^{\pi/2} + (n-1) \int_0^{\pi/2} \sin^{n-2} x \cos^2 x \, dx$$

The boundary term vanishes (both endpoints give zero). Use $\cos^2 x = 1 - \sin^2 x$:

$$I_n = (n-1)(I_{n-2} - I_n) \Rightarrow nI_n = (n-1)I_{n-2} \Rightarrow I_n = \frac{n-1}{n} I_{n-2}.$$

Hence find $\int_0^{\pi/2} \sin^6 x \, dx$.

$$I_0 = \pi/2. \quad I_2 = \frac{1}{2}I_0 = \pi/4. \quad I_4 = \frac{3}{4}I_2 = 3\pi/16. \quad I_6 = \frac{5}{6}I_4 = \frac{15\pi}{96} = \frac{5\pi}{32}.$$

OCR Exam Tip: State the substitution or choice of u and dv clearly. When the boundary terms vanish, say so and give the values — don't just drop them silently.

6. Arc Length

The length of the curve $y = f(x)$ from $x = a$ to $x = b$ is

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx.$$

For parametric curves $x = x(t)$, $y = y(t)$:

$$L = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt.$$

Arc length of $y = \frac{2}{3}x^{3/2}$ from $x = 0$ to $x = 3$.

$\frac{dy}{dx} = x^{1/2}$, so integrand = $\sqrt{1+x}$.

$$L = \int_0^3 \sqrt{1+x} dx = \left[\frac{2}{3}(1+x)^{3/2}\right]_0^3 = \frac{2}{3}(8-1) = \frac{14}{3}.$$

7. Surface Area of Revolution

Rotating $y = f(x)$, $a \leq x \leq b$, about the x -axis produces a surface of area

$$S = 2\pi \int_a^b y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx.$$

About the y -axis (if $x \geq 0$ throughout):

$$S = 2\pi \int_a^b x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx.$$

Parametric form: replace $\sqrt{1 + (dy/dx)^2} dx$ by $\sqrt{(dx/dt)^2 + (dy/dt)^2} dt$ and y (or x) by its parametric expression.

8. Polar and Parametric Forms (Support)

Polar arc length and area are covered in §1.7; here we note that calculus on curves in any representation comes down to arc length element ds :

$$ds^2 = dx^2 + dy^2$$

9. Exam Style Strategy

- For improper integrals, show the limit explicitly.
- For reduction formulae, establish the base case(s) I_0 or I_1 cleanly.
- Arc length and surface integrals are sensitive to the *axis* of rotation — reread the question.
- Trig / hyperbolic substitutions: write down both the substitution and dx ; update limits if definite.